

$$12. \left[\begin{array}{ccccc|c} 1 & -1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

In Exercises 13–20, find all solutions of the given linear system, using the Gauss method with back substitution.

13. $2x - y = 8$

$6x - 5y = 32$

14. $4x_1 - 3x_2 = 10$

$8x_1 - x_2 = 10$

15. $y + z = 6$

$3x - y + z = -7$

$x + y - 3z = -13$

16. $2x + y - 3z = 0$

$6x + 3y - 8z = 0$

$2x - y + 5z = -4$

17. $x_1 - 2x_2 = 3$

$3x_1 - x_2 = 14$

$x_1 - 7x_2 = -2$

18. $x_1 - 3x_2 + x_3 = 2$

$3x_1 - 8x_2 + 2x_3 = 5$

19. $x_1 + 4x_2 - 2x_3 = 4$

$2x_1 + 7x_2 - x_3 = -2$

$2x_1 + 9x_2 - 7x_3 = 1$

20. $x_1 - 3x_2 + 2x_3 - x_4 = 8$

$3x_1 - 7x_2 + x_4 = 0$

In Exercises 21–24, find all solutions of the linear system, using the Gauss–Jordan method.

21. $3x_1 - 2x_2 = -8$

$4x_1 + 5x_2 = -3$

22. $2x_1 + 8x_2 = 16$

$5x_1 - 4x_2 = -8$

23. $x_1 - 2x_3 + x_4 = 6$

$2x_1 - x_2 + x_3 - 3x_4 = 0$

$9x_1 - 3x_2 - x_3 - 7x_4 = 4$

24. $x_1 + 2x_2 - 3x_3 + x_4 = 2$

$3x_1 + 6x_2 - 8x_3 - 2x_4 = 1$

In Exercises 25–28, determine whether the vector $\mathbf{b}$ is in the span of the vectors $\mathbf{v}_i$.

25. $\mathbf{b} = \begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} -3 \\ -1 \\ 5 \end{bmatrix}$

26. $\mathbf{b} = \begin{bmatrix} 8 \\ 26 \\ 14 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -4 \\ -12 \\ -9 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$

27. $\mathbf{b} = \begin{bmatrix} 8 \\ 17 \\ -8 \\ 3 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 5 \\ -2 \\ 5 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} -3 \\ -6 \\ 1 \\ -8 \end{bmatrix}$,
 $\mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ -4 \end{bmatrix}$

28. $\mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 7 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -3 \\ -2 \\ -8 \\ -9 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 4 \end{bmatrix}$,
 $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}$

29. Mark each of the following True or False.

- a. Every linear system with the same number of equations as unknowns has a unique solution.
- b. Every linear system with the same number of equations as unknowns has at least one solution.
- c. A linear system with more equations than unknowns may have an infinite number of solutions.
- d. A linear system with fewer equations than unknowns may have no solution.
- e. Every matrix is row equivalent to a unique matrix in row-echelon form.
- f. Every matrix is row equivalent to a unique matrix in reduced row-echelon form.
- g. If $[A \mid \mathbf{b}]$ and $[B \mid \mathbf{c}]$ are row-equivalent partitioned matrices, the linear systems

$Ax = b$ and $Bx = c$ have the same solution set.

- h. A linear system with a square coefficient matrix A has a unique solution if and only if A is row equivalent to the identity matrix.
- i. A linear system with coefficient matrix A has an infinite number of solutions if and only if A can be row-reduced to an echelon matrix that includes some column containing no pivot.
- j. A consistent linear system with coefficient matrix A has an infinite number of solutions if and only if A can be row-reduced to an echelon matrix that includes some column containing no pivot.

In Exercises 30–37, describe all possible values for the unknowns x_i so that the matrix equation is valid.

30. $2[x_1 \ x_2] - [4 \ 7] = [-2 \ 11]$

31. $4[x_1 \ x_2] + 2[x_1 \ 3] = [-6 \ 18]$

32. $[x_1 \ x_2] \begin{bmatrix} 1 \\ -3 \end{bmatrix} = [2]$

33. $[x_1 \ x_2] \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} = [0 \ -14]$

34. $\begin{bmatrix} 1 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$

35. $\begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 3 & 1 \end{bmatrix}$

36. $[x_1 \ x_2] \begin{bmatrix} 3 & 0 & 4 \\ 2 & 1 & -1 \end{bmatrix} = [3 \ 3 \ -7]$

37. $\begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

38. Determine all values of the b_i that make the linear system

$$x_1 + 2x_2 = b_1$$

$$3x_1 + 6x_2 = b_2$$

consistent.

39. Determine all values b_1 and b_2 such that $b = [b_1, b_2]$ is a linear combination of $v_1 = [1, 3]$ and $v_2 = [5, -1]$.

40. Determine all values of the b_i that make the linear system

$$x_1 + x_2 - x_3 = b_1$$

$$2x_2 + x_3 = b_2$$

$$x_2 - x_3 = b_3$$

consistent.

41. Determine all values $b_1, b_2,$ and b_3 such that $b = [b_1, b_2, b_3]$ lies in the span of $v_1 = [1, 1, 0]$, $v_2 = [3, -1, 4]$, and $v_3 = [-1, 2, -3]$.

42. Find an elementary matrix E such that

$$E \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 3 & 4 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & 2 & -11 \end{bmatrix}.$$

43. Find an elementary matrix E such that

$$E \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 3 & 4 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 2 & 7 & 4 & 9 \\ 3 & 4 & 5 & 1 \end{bmatrix}.$$

44. Find a matrix C such that

$$C \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \\ 0 & -6 \end{bmatrix}.$$

45. Find a matrix C such that

$$C \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & 2 \\ 1 & 2 \end{bmatrix}.$$

In Exercises 46–51, let A be a 4×4 matrix. Find a matrix C such that the result of applying the given sequence of elementary row operations to A can also be found by computing the product CA .

46. Interchange row 1 and row 2.
47. Interchange row 1 and row 3; multiply row 3 by 4.
48. Multiply row 1 by 5; interchange rows 2 and 3; add 2 times row 3 to row 4.
49. Add 4 times row 2 to row 4; multiply row 4 by -3 ; add 5 times row 4 to row 1.
50. Interchange rows 1 and 4; add 6 times row 2 to row 1; add -3 times row 1 to row 3; add -2 times row 4 to row 2.

51. Add 3 times row 2 to row 4; add -2 times row 4 to row 3; add 5 times row 3 to row 1; add -4 times row 1 to row 2.

Exercise 24 in Section 1.3 is useful for the next three exercises.

52. Prove Theorem 1.8 for the row-interchange operation.
53. Prove Theorem 1.8 for the row-scaling operation.
54. Prove Theorem 1.8 for the row-addition operation.
55. Prove that row equivalence $\sim$ is an equivalence relation by verifying the following for $m \times n$ matrices A , B , and C .
- $A \sim A$. (Reflexive Property)
 - If $A \sim B$, then $B \sim A$. (Symmetric Property)
 - If $A \sim B$ and $B \sim C$, then $A \sim C$. (Transitive Property)
56. Find a , b , and c such that the parabola $y = ax^2 + bx + c$ passes through the points $(1, -4)$, $(-1, 0)$, and $(2, 3)$.
57. Find a , b , c , and d such that the quartic curve $y = ax^4 + bx^3 + cx^2 + d$ passes through $(1, 2)$, $(-1, 6)$, $(-2, 38)$, and $(2, 6)$.
58. Let A be an $m \times n$ matrix, and let $\mathbf{c}$ be a column vector such that $A\mathbf{x} = \mathbf{c}$ has a unique solution.
- Prove that $m \geq n$.
 - If $m = n$, must the system $A\mathbf{x} = \mathbf{b}$ be consistent for every choice of $\mathbf{b}$?
 - Answer part (b) for the case where $m > n$.

 A problem we meet when reducing a matrix with the aid of a computer involves determining when a computed entry should be 0. The computer might give an entry as 0.00000001, because of roundoff error, when it should be 0. If the computer uses this entry as a pivot in a future step, the result is chaotic! For this reason, it is common practice to program the computer to replace all sufficiently small computed entries with

0, where the meaning of "sufficiently small" must be specified in terms of the size of the nonzero entries in the original matrix. The routine YUREDUCE in LINTEK provides drill on the steps involved in reducing a matrix without requiring burdensome computation. The program computes the smallest nonzero coefficient magnitude m and asks the user to enter a number r (for ratio); all computed entries of magnitude less than rm produced during reduction of the coefficient matrix will be set equal to zero. In Exercises 59–64, use the routine YUREDUCE, specifying $r = 0.0001$, to solve the linear system.

59. $3x_1 - x_2 = -10$
 $7x_1 + 2x_2 = 7$
 $2x_1 - 5x_2 = -37$
60. $5x_1 - 2x_2 = 11$
 $8x_1 + x_2 = 3$
 $6x_1 - 5x_2 = -4$
61. $7x_1 - 2x_2 + x_3 = -14$
 $-4x_1 + 5x_2 - 3x_3 = 17$
 $5x_1 - x_2 + 2x_3 = -7$
62. $-3x_1 + 5x_2 + 2x_3 = 12$
 $5x_1 - 7x_2 + 6x_3 = -16$
 $11x_1 - 17x_2 + 2x_3 = -40$
63. $x_1 - 2x_2 + x_3 - x_4 + 2x_5 = 1$
 $2x_1 + x_2 - 4x_3 - x_4 + 5x_5 = 16$
 $8x_1 - x_2 + 3x_3 - x_4 - x_5 = 1$
 $4x_1 - 2x_2 + 3x_3 - 8x_4 + 2x_5 = -5$
 $5x_1 + 3x_2 - 4x_3 + 7x_4 - 6x_5 = 7$
64. $x_1 - 2x_2 + x_3 - x_4 + 2x_5 = 1$
 $2x_1 + x_2 - 4x_3 - x_4 + 5x_5 = 10$
 $8x_1 - x_2 + 3x_3 - x_4 - x_5 = -5$
 $4x_1 - 2x_2 + 3x_3 - 8x_4 + 2x_5 = -3$
 $5x_1 + 3x_2 - 4x_3 + 7x_4 - 6x_5 = 1$

The routine MATCOMP in LINTEK can also be used to find the solutions of a linear system. MATCOMP will bring the left portion of the augmented matrix to reduced row-echelon form and display the result on the screen. The user can

then find the solutions. Use *MATCOMP* in the remaining exercises.

65. Find the reduced row-echelon form of the matrix in Exercise 6, by taking it as a coefficient matrix for zero systems.

66. Solve the linear system in Exercise 61.
67. Solve the linear system in Exercise 62.
68. Solve the linear system in Exercise 63.

MATLAB

When reducing a matrix X to reduced row-echelon form, we may need to swap row i with row k . This can be done in MATLAB using the command

$$X([i \ k],:) = X([k \ i],:).$$

If we wish to multiply the i th row by the reciprocal of x_{ij} to create a pivot 1 in the i th row and j th column, we can give the command

$$X(i,:) = X(i,)/X(i,j).$$

When we have made pivots 1 and wish to make the entry in row k , column j equal to zero using the pivot in row i , column j , we always multiply row i by the negative of the entry that we wish to make zero, and add the result to row k . In MATLAB, this has the form

$$X(k,:) = X(k,:) - X(k,j)*X(i,:).$$

Access MATLAB and enter the lines

```
X = ones(4); i = 1; j = 2; k = 3;
X([i k],:) = X([k i],:)
X(i,:) = X(i,)/X(i,j)
X(k,:) = X(k,:) - X(k,j)*X(i,);
```

which you can then access using the up-arrow key and edit repeatedly to row-reduce a matrix X . MATLAB will not show a partition in X —you have to supply the partition mentally. If your installation contains the data files for our text, enter **flc1s4** now. We will be asking you to work with some of the augmented matrices used in the exercises for this section. In our data file, the augmented matrix for Exercise 63 is called **E63**, etc. Solve the indicated system by setting X equal to the appropriate matrix and reducing it using the up-arrow key and editing repeatedly the three basic commands above. In MATLAB, only the commands executed most recently can be accessed by using the up-arrow key. To avoid losing the command to interchange rows, which is seldom necessary, execute it at least once in each exercise even if it is not needed. (Interchanging the same rows twice leaves a matrix unchanged.) Solve the indicated exercises listed below.

- M1. Exercise 21
M2. Exercise 23
M3. Exercise 60

- M4. Exercise 61
M5. Exercise 62

The command `rref(A)` in MATLAB will reduce the matrix A to reduced row-echelon form. Use this command to solve the following exercises.

M6. Exercise 6

M8. Exercise 63

M7. Exercise 24

M9. Exercise 64

(MATLAB contains a demo command `rrefmovie(A)` designed to show the step-by-step reduction of A , but with our copy and a moderately fast computer, the demo goes so fast that it is hard to catch it with the Pause key in order to view it. If you are handy with a word processor, you might copy the file `rrefmovi.m` as `rrefmovp.m`, and then edit `rrefmovp.m` to supply `,pause` at the end of each of the four lines that start with `A(`. Entering `rrefmovp(A)` from MATLAB will then run the altered demo, which will pause after each change of a matrix. Strike any key to continue after a pause. You may notice that there seems to be unnecessary row swapping to create pivots. Look at the paragraph on partial pivoting in Section 10.3 to understand the reason for this.)

1.5

INVERSES OF SQUARE MATRICES

Matrix Equations and Inverses

A system of n equations in n unknowns $x_1, x_2, \dots, x_n$ can be expressed in matrix form as

$$Ax = b, \quad (1)$$

where A is the $n \times n$ coefficient matrix, x is the $n \times 1$ column vector with i th entry x_i , and b is an $n \times 1$ column vector with constant entries. The analogous equation using scalars is

$$ax = b \quad (2)$$

for scalars a and b . If $a \neq 0$, we usually think of solving Eq. (2) for x by dividing by a , but we can just as well think of solving it by multiplying by $1/a$. Breaking the solution down into small steps, we have

$$\begin{aligned} \left(\frac{1}{a}\right)ax &= \left(\frac{1}{a}\right)b && \text{Multiplication by } 1/a \\ \left[\left(\frac{1}{a}\right)a\right]x &= \left(\frac{1}{a}\right)b && \text{Associativity of multiplication} \\ 1x &= \left(\frac{1}{a}\right)b && \text{Property of } 1/a \\ x &= \left(\frac{1}{a}\right)b. && \text{Property of } 1 \end{aligned}$$

Let us see whether we can solve Eq. (1) similarly if A is a nonzero matrix. Matrix multiplication is associative, and the $n \times n$ identity matrix I plays the same role for multiplication of $n \times n$ matrices that the number 1 plays for

multiplication of numbers. The crucial step is to find an $n \times n$ matrix C such that $CA = I$, so that C plays for matrices the role that $1/a$ does for numbers. If such a matrix C exists, we can obtain from Eq. (1)

$$C(Ax) = Cb \quad \text{Multiplication by } C$$

$$(CA)x = Cb \quad \text{Associativity of multiplication}$$

$$Ix = Cb \quad \text{Property of } C$$

$$x = Cb, \quad \text{Property of } I$$

which shows that our column vector x of unknowns must be the column vector Cb . Now an interesting problem arises. When we substitute $x = Cb$ back into our equation $Ax = b$ to verify that we do indeed have a solution, we obtain $Ax = A(Cb) = (AC)b$. But how do we know that $AC = I$ from our assumption that $CA = I$? Matrix multiplication is not a commutative operation. This problem does not arise with our scalar equation $ax = b$, because multiplication of real numbers is commutative. It is indeed true that for square matrices, if $CA = I$ then $AC = I$, and we will work toward a proof of this. For example, the equation

$$\begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix}$$

illustrates that

$$A = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix}$$

satisfy $CA = I = AC$.

Unfortunately, it is not true that, for each nonzero $n \times n$ matrix A , we can find an $n \times n$ matrix C such that $CA = AC = I$. For example, if the first column of A has only zero entries, then the first column of CA also has only zero entries for any matrix C , so $CA \neq I$ for any matrix C . However, for many important $n \times n$ matrices A , there does exist an $n \times n$ matrix C such that $CA = AC = I$. Let us show that when such a matrix exists, it is unique.

THEOREM 1.9 Uniqueness of an Inverse

Let A be an $n \times n$ matrix. If C and D are matrices such that $AC = DA = I$, then $C = D$. In particular, if $AC = CA = I$, then C is the *unique* matrix with this property.

PROOF Let C and D be matrices such that $AC = DA = I$. Because matrix multiplication is associative, we have

$$D(AC) = (DA)C.$$

But, because $AC = I$ and $DA = I$, we find that

$$D(AC) = DI = D \quad \text{and} \quad (DA)C = IC = C.$$

Therefore, $C = D$.

Now suppose that $AC = CA = I$, and let us show that C is the unique matrix with this property. To this end, suppose also that $AD = DA = I$. Then we have $AC = I = DA$, so $D = C$, as we just showed. $\blacktriangle$

From the title of the preceding theorem, we anticipate the following definition.

DEFINITION 1.15 Invertible Matrix

An $n \times n$ matrix A is **invertible** if there exists an $n \times n$ matrix C such that $CA = AC = I$, the $n \times n$ identity matrix. The matrix C is the inverse of A and is denoted by A^{-1} . If A is not invertible, it is **singular**.

Although A^{-1} plays the same role arithmetically as $a^{-1} = 1/a$ (as we showed at the start of this section), we will never write A^{-1} as $1/A$. The powers of an invertible $n \times n$ matrix A are now defined for all integers. That is, for $m > 0$, A^m is the product of m factors A , and A^{-m} is the product of m factors A^{-1} . We consider A^0 to be the $n \times n$ identity matrix I .

Inverses of Elementary Matrices

In Section 1.4, we saw that each elementary row operation can be undone by another (possibly the same) elementary row operation. Let us see how this fact

HISTORICAL NOTE THE NOTION OF THE INVERSE OF A MATRIX first appears in an 1855 note of Arthur Cayley (1821–1895) and is made more explicit in an 1858 paper entitled “A Memoir on the Theory of Matrices.” In that work, Cayley outlines the basic properties of matrices, noting that most of these derive from work with sets of linear equations. In particular, the inverse comes from the idea of solving a system

$$X = ax + by + cz$$

$$Y = a'x + b'y + c'z$$

$$Z = a''x + b''y + c''z$$

for x, y, z in terms of X, Y, Z . Cayley gives an explicit construction for the inverse in terms of the determinants of the original matrix and of the minors.

In 1842, Arthur Cayley graduated from Trinity College, Cambridge, but could not find a suitable teaching post. So, like Sylvester, he studied law and was called to the bar in 1849. During his 14 years as a lawyer, he wrote about 300 mathematical papers; finally, in 1863 he became a professor at Cambridge, where he remained until his death. It was during his stint as a lawyer that he met Sylvester; their discussions over the next 40 years were extremely fruitful for the progress of algebra. Over his lifetime, Cayley produced about 1000 papers in pure mathematics, theoretical dynamics, and mathematical astronomy.

translates to the invertibility of the elementary matrices, and a description of their inverses.

Let E_1 be an elementary row-interchange matrix, obtained from the identity matrix I by the interchanging of rows i and k . Recall that $E_1 A$ effects the interchange of rows i and k of A for any matrix A such that $E_1 A$ is defined. In particular, taking $A = E_1$, we see that $E_1 E_1$ interchanges rows i and k of E_1 , and hence changes E_1 back to I . Thus,

$$E_1 E_1 = I.$$

Consequently, E_1 is an invertible matrix and is its own inverse.

Now let E_2 be an elementary row-scaling matrix, obtained from the identity matrix by the multiplication of row i by a nonzero scalar r . Let E_2' be the matrix obtained from the identity matrix by the multiplication of row i by $1/r$. It is clear that

$$E_2' E_2 = E_2 E_2' = I,$$

so E_2 is invertible, with inverse E_2' .

Finally, let E_3 be an elementary row-addition matrix, obtained from I by the addition of r times row i to row k . If E_3' is obtained from I by the addition of $-r$ times row i to row k , then

$$E_3' E_3 = E_3 E_3' = I.$$

We have established the following fact:

Every elementary matrix is invertible.

EXAMPLE 1 Find the inverses of the elementary matrices

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{and} \quad E_3 = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

SOLUTION Because E_1 is obtained from

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by the interchanging of the first and second rows, we see that $E_1^{-1} = E_1$.

The matrix E_2 is obtained from I by the multiplication of the first row by 3, so we must multiply the first row of I by $\frac{1}{3}$ to form

$$E_2^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Finally, E_3 is obtained from I by the addition of 4 times row 3 to row 1. To form E_3^{-1} from I , we add -4 times row 3 to row 1, so

$$E_3^{-1} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Inverses of Products

The next theorem is fundamental in work with inverses.

THEOREM 1.10 Inverses of Products

Let A and B be invertible $n \times n$ matrices. Then AB is invertible, and $(AB)^{-1} = B^{-1}A^{-1}$.

PROOF By assumption, there exist matrices A^{-1} and B^{-1} such that $AA^{-1} = A^{-1}A = I$ and $BB^{-1} = B^{-1}B = I$. Making use of the associative law for matrix multiplication, we find that

$$(AB)(B^{-1}A^{-1}) = [A(BB^{-1})]A^{-1} = (AI)A^{-1} = AA^{-1} = I.$$

A similar computation shows that $(B^{-1}A^{-1})(AB) = I$. Therefore, the inverse of AB is $B^{-1}A^{-1}$; that is, $(AB)^{-1} = B^{-1}A^{-1}$. ▲

It is instructive to apply Theorem 1.10 to a product $E_t \cdots E_3E_2E_1$ of elementary matrices. In the expression

$$(E_t \cdots E_3E_2E_1)A,$$

the product $E_t \cdots E_3E_2E_1$ performs a sequence of elementary row operations on A . First, E_1 acts on A ; then, E_2 acts on E_1A ; and so on. To undo this sequence, we must first undo the last elementary row operation, performed by E_t . This is accomplished by using E_t^{-1} . Continuing, we should perform the sequence of operations given by

$$E_1^{-1}E_2^{-1}E_3^{-1} \cdots E_t^{-1}$$

in order to effect $(E_t \cdots E_3E_2E_1)^{-1}$.

A Commutativity Property

We are now in position to show that if $CA = I$, then $AC = I$. First we prove a lemma (a result preliminary to the main result).

LEMMA 1.1 Condition for $Ax = b$ to Be Solvable for All b

Let A be an $n \times n$ matrix. The linear system $Ax = b$ has a solution for every choice of column vector $b \in \mathbb{R}^n$ if and only if A is row equivalent to the $n \times n$ identity matrix I .

PROOF Let b be any column vector in $\mathbb{R}^n$ and let the augmented matrix $[A \mid b]$ be row-reduced to $[H \mid c]$ where H is in reduced row-echelon form. If H is the identity matrix I , then the linear system $Ax = b$ has the solution $x = c$.

For the converse, suppose that reduction of A to reduced row-echelon form yields a matrix H that is not the identity matrix. Then the bottom row of H must have every entry equal to 0. Now there exist elementary matrices $E_1, E_2, \dots, E_t$ such that $(E_t \cdots E_2 E_1)A = H$. Recall that every elementary matrix is invertible, and that a product of elementary matrices is invertible. Let $b = (E_t \cdots E_2 E_1)^{-1} e_n$, where e_n is the column vector with 1 in its n th component and zeros elsewhere. Reduction of the augmented matrix $[A \mid b]$ can be accomplished by multiplying both A and b on the left by $E_t \cdots E_2 E_1$, so the reduction will yield $[H \mid e_n]$, which represents a system of equations with no solution because the bottom row has entries 0 to the left of the partition and 1 to the right of the partition. This shows that if H is not the identity matrix, then $Ax = b$ does not have a solution for some $b \in \mathbb{R}^n$. $\blacktriangle$

THEOREM 1.11 A Commutativity Property

Let A and C be $n \times n$ matrices. Then $CA = I$ if and only if $AC = I$.

PROOF To prove that $CA = I$ if and only if $AC = I$, it suffices to prove that if $AC = I$, then $CA = I$, because the converse statement is obtained by reversing the roles of A and C .

Suppose now that we do have $AC = I$. Then the equation $Ax = b$ has a solution for every column vector b in $\mathbb{R}^n$; we need only notice that $x = Cb$ is a solution because $A(Cb) = (AC)b = Ib = b$. By Lemma 1.1, we know that A is row equivalent to the $n \times n$ identity matrix I , so there exists a sequence of elementary matrices $E_1, E_2, \dots, E_t$ such that $(E_t \cdots E_2 E_1)A = I$. By Theorem 1.9, the two equations

$$(E_t \cdots E_2 E_1)A = I \quad \text{and} \quad AC = I$$

imply that $E_t \cdots E_2 E_1 = C$, so we have $CA = I$ also. $\blacktriangle$

Computation of Inverses

Let $A = [a_{ij}]$ be an $n \times n$ matrix. To find A^{-1} , if it exists, we must find an $n \times n$ matrix $X = [x_{ij}]$ such that $AX = I$ —that is, such that

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}. \quad (3)$$

Matrix equation (3) corresponds to n^2 linear equations in the n^2 unknowns x_{ij} ; there is one linear equation for each of the n^2 positions in an $n \times n$ matrix. For example, equating the entries in the row 2, column 1 position on each side of Eq. (3), we obtain the linear equation

$$a_{21}x_{11} + a_{22}x_{21} + \cdots + a_{2n}x_{n1} = 0.$$

Of these n^2 linear equations, n of them involve the n unknowns x_{i1} for $i = 1, 2, \dots, n$; and these equations are given by the column-vector equation

$$A \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad (4)$$

which is a square system of equations. There are also n equations involving the n unknowns x_{i2} , for $i = 1, 2, \dots, n$; and so on. In addition to solving system (4), we must solve the systems

$$A \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, A \begin{bmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}, \quad (5)$$

where each system has the same coefficient matrix A . Whenever we want to solve several systems $A\mathbf{x} = \mathbf{b}_i$ with the same coefficient matrix but different vectors $\mathbf{b}_i$, we solve them all at once, rather than one at a time. The main job in solving a linear system is reducing the coefficient matrix to row-echelon or reduced row-echelon form, and we don't want to repeat that work over and over. We simply reduce one augmented matrix, where we line up all the vectors $\mathbf{b}_i$ to the right of the partition. Thus, to solve all the linear systems in Eqs. (4) and (5), we form the augmented matrix

$$\left[\begin{array}{cccc|cccc} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & 0 & 0 & \cdots & 1 \end{array} \right], \quad (6)$$

which we abbreviate by $[A | I]$. The matrix A is to the left of the partition, and the identity matrix I is to the right. We then perform a Gauss–Jordan reduction on this augmented matrix. By Theorem 1.9, we know that if A^{-1} exists, it is *unique*, so that every column in the reduced row-echelon form of A has a pivot. Thus, A^{-1} exists if and only if the augmented matrix (6) can be reduced to

$$\left[\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & c_{11} & c_{12} & \cdots & c_{1n} \\ 0 & 1 & \cdots & 0 & c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & c_{n1} & c_{n2} & \cdots & c_{nn} \end{array} \right],$$

where the $n \times n$ identity matrix I is to the left of the partition. The $n \times n$ solution matrix $C = [c_{ij}]$ to the right of the partition then satisfies $AC = I$, so $A^{-1} = C$. This is an efficient way to compute A^{-1} . We summarize the computation in the following box, and we state the theory in Theorem 1.12.

Computation of A^{-1}

To find A^{-1} , if it exists, proceed as follows:

Step 1 Form the augmented matrix $[A | I]$.

Step 2 Apply the Gauss–Jordan method to attempt to reduce $[A | I]$ to $[I | C]$. If the reduction can be carried out, then $A^{-1} = C$. Otherwise, A^{-1} does not exist.

EXAMPLE 2 For the matrix $A = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix}$, compute the inverse we exhibited at the start of this section, and use this inverse to solve the linear system

$$\begin{aligned} 2x + 9y &= -5 \\ x + 4y &= 7. \end{aligned}$$

SOLUTION Reducing the augmented matrix, we have

$$\begin{aligned} \left[\begin{array}{cc|cc} 2 & 9 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{array} \right] &\sim \left[\begin{array}{cc|cc} 1 & 4 & 0 & 1 \\ 2 & 9 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 4 & 0 & 1 \\ 0 & 1 & 1 & -2 \end{array} \right] \\ &\sim \left[\begin{array}{cc|cc} 1 & 0 & -4 & 9 \\ 0 & 1 & 1 & -2 \end{array} \right]. \end{aligned}$$

Therefore,

$$A^{-1} = \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix}.$$

If A^{-1} exists, the solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$. Consequently, the solution of our system

$$\begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5 \\ 7 \end{bmatrix} \quad \text{is} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} -5 \\ 7 \end{bmatrix} = \begin{bmatrix} 83 \\ -19 \end{bmatrix}.$$

We emphasize that the computation of the solution of the linear system in Example 2, using the inverse of the coefficient matrix, was for illustration only. When faced with the problem of solving a square system, $A\mathbf{x} = \mathbf{b}$, one should never start by finding the inverse of the coefficient matrix. To do so would involve row reduction of $[A \mid I]$ and subsequent computation of $A^{-1}\mathbf{b}$, whereas the shorter reduction of $[A \mid \mathbf{b}]$ provides the desired solution at once. The inverse of a matrix is often useful in symbolic computations. For example, if A is an invertible matrix and we know that $AB = AC$, then we can deduce that $B = C$ by multiplying both sides of $AB = AC$ on the left by A^{-1} . If we have r systems of equations

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2, \quad \dots, \quad A\mathbf{x}_r = \mathbf{b}_r,$$

all with the same invertible $n \times n$ coefficient matrix A , it might seem to be more efficient (for large r) to solve all the systems by finding A^{-1} and computing the column vectors

$$\mathbf{x}_1 = A^{-1}\mathbf{b}_1, \quad \mathbf{x}_2 = A^{-1}\mathbf{b}_2, \quad \dots, \quad \mathbf{x}_r = A^{-1}\mathbf{b}_r.$$

Section 10.1 will show that using the Gauss method with back substitution on the augmented matrix $[A \mid \mathbf{b}_1 \ \mathbf{b}_2 \ \dots \ \mathbf{b}_r]$ remains more efficient. Thus, inversion of a coefficient matrix is not a good numerical way to solve a linear system. However, we will find inverses very useful for solving other kinds of problems.

THEOREM 1.12 Conditions for A^{-1} to Exist

The following conditions for an $n \times n$ matrix A are equivalent:

- (i) A is invertible.
- (ii) A is row equivalent to the identity matrix I .
- (iii) The system $A\mathbf{x} = \mathbf{b}$ has a solution for each n -component column vector $\mathbf{b}$.
- (iv) A can be expressed as a product of elementary matrices.
- (v) The span of the column vectors of A is $\mathbb{R}^n$.

PROOF Step 2 in the box preceding Example 2 shows that parts (i) and (ii) of Theorem 1.12 are equivalent. For the equivalence of (ii) with (iii), Lemma 1.1 shows that $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b} \in \mathbb{R}^n$ if and only if (ii) is true. Thus, (ii) and (iii) are equivalent. The equivalence of (iii) and (v) follows from the box on page 63.

Turning to the equivalence of parts (ii) and (iv), we know that the matrix A is row equivalent to I if and only if there is a sequence of elementary matrices $E_1, E_2, \dots, E_t$ such that $E_t \cdots E_2 E_1 A = I$; and this is the case if and only if A is expressible as a product $A = E_1^{-1} E_2^{-1} \cdots E_t^{-1}$ of elementary matrices. $\blacktriangle$

EXAMPLE 3 Using Example 2, express $\begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix}$ as a product of elementary matrices.

SOLUTION The steps we performed in Example 2 can be applied in sequence to 2×2 identity matrices to generate elementary matrices:

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{Interchange rows 1 and 2.}$$

$$E_2 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \quad \text{Add } -2 \text{ times row 1 to row 2.}$$

$$E_3 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}. \quad \text{Add } -4 \text{ times row 2 to row 1.}$$

Thus we see that $E_3 E_2 E_1 A = I$, so

$$A = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}. \quad \blacksquare$$

Example 3 illustrates the following boxed rule for expressing an invertible matrix A as a product of elementary matrices.

Expressing an Invertible Matrix A as a Product of Elementary Matrices

Write in left-to-right order the inverses of the elementary matrices corresponding to successive row operations that reduce A to I .

EXAMPLE 4 Determine whether the matrix

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 5 & -3 \\ -3 & 2 & -4 \end{bmatrix}$$

is invertible, and find its inverse if it is.

SOLUTION We have

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & 3 & -2 & 1 & 0 & 0 \\ 2 & 5 & -3 & 0 & 1 & 0 \\ -3 & 2 & -4 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 3 & -2 & 1 & 0 & 0 \\ 0 & -1 & 1 & -2 & 1 & 0 \\ 0 & 11 & -10 & 3 & 0 & 1 \end{array} \right] \\ & \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -5 & 3 & 0 \\ 0 & 1 & -1 & 2 & -1 & 0 \\ 0 & 0 & 1 & -19 & 11 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 14 & -8 & -1 \\ 0 & 1 & 0 & -17 & 10 & 1 \\ 0 & 0 & 1 & -19 & 11 & 1 \end{array} \right]. \end{aligned}$$

Therefore, A is an invertible matrix, and

$$A^{-1} = \begin{bmatrix} 14 & -8 & -1 \\ -17 & 10 & 1 \\ -19 & 11 & 1 \end{bmatrix}.$$

EXAMPLE 5 Express the matrix A of Example 4 as a product of elementary matrices.

SOLUTION In accordance with the box that follows Example 3, we write in left-to-right order the successive inverses of the elementary matrices corresponding to the row reduction of A in Example 4. We obtain

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

EXAMPLE 6 Determine whether the span of the vectors $[1, -2, 1]$, $[3, -5, 4]$, and $[4, -3, 9]$ is all of $\mathbb{R}^3$.

SOLUTION Let

$$A = \begin{bmatrix} 1 & 3 & 4 \\ -2 & -5 & -3 \\ 1 & 4 & 9 \end{bmatrix}.$$

We have

$$\begin{bmatrix} 1 & 3 & 4 \\ -2 & -5 & -3 \\ 1 & 4 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 5 \\ 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}.$$

We do not have a pivot in the row 3, column 3 position, so we are not able to reduce A to the identity matrix. By Theorem 1.12, the span of the given vectors is not all of $\mathbb{R}^3$. ■

SUMMARY

1. Let A be a square matrix. A square matrix C such that $CA = AC = I$ is the inverse of A and is denoted by $C = A^{-1}$. If such an inverse of A exists, then A is said to be *invertible*. The inverse of an invertible matrix A is unique. A square matrix that has no inverse is called *singular*.
2. The inverse of a square matrix A exists if and only if A can be reduced to the identity matrix I by means of elementary row operations or (equivalently) if and only if A is a product of elementary matrices. In this case, A is equal to the product, in left-to-right order, of the inverses of the successive

elementary matrices corresponding to the sequence of row operations used to reduce A to I .

- To find A^{-1} , if it exists, form the augmented matrix $[A \mid I]$ and apply the Gauss–Jordan method to reduce this matrix to $[I \mid C]$. If this can be done, then $A^{-1} = C$. Otherwise, A is not invertible.
- The inverse of a product of invertible matrices is the product of the inverses in the reverse order.

EXERCISES

In Exercises 1–8, (a) find the inverse of the square matrix, if it exists, and (b) express each invertible matrix as a product of elementary matrices.

1. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 3 & 6 \\ 4 & 8 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

7. $\begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 5 \\ 0 & -1 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 3 & 6 \\ 3 & 8 \end{bmatrix}$

4. $\begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ -3 & 1 & -7 \end{bmatrix}$

8. $\begin{bmatrix} -1 & 2 & 1 \\ 2 & -3 & 5 \\ 1 & 0 & 12 \end{bmatrix}$

In Exercises 11 and 12, determine whether the span of the column vectors of the given matrix is $\mathbb{R}^4$.

11. $\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & -1 & -3 & 4 \\ 1 & 0 & -1 & 2 \\ -3 & 0 & 0 & -1 \end{bmatrix}$

12. $\begin{bmatrix} 1 & -2 & 1 & 0 \\ -3 & 5 & 0 & 2 \\ 0 & 1 & 2 & -4 \\ -1 & 2 & 4 & -2 \end{bmatrix}$

13. a. Show that the matrix

$$A = \begin{bmatrix} 2 & -3 \\ 5 & -7 \end{bmatrix}$$

is invertible, and find its inverse.

- b. Use the result in (a) to find the solution of the system of equations

$$2x_1 - 3x_2 = 4, \quad 5x_1 - 7x_2 = -3.$$

14. Using the inverse of the matrix in Exercise 7, find the solution of the system of equations

$$2x_1 + x_2 + 4x_3 = 5$$

$$3x_1 + 2x_2 + 5x_3 = 3$$

$$-x_2 + x_3 = 8.$$

15. Find three linear equations that express x , y , z in terms of r , s , t , if

$$2x + y + 4z = r$$

$$3x + 2y + 5z = s$$

$$-y + z = t.$$

[HINT: See Exercise 14.]

In Exercises 9 and 10, find the inverse of the matrix, if it exists.

9. $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix}$

10. $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

16. Let

$$A^{-1} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 4 & 1 & 2 \end{bmatrix}.$$

If possible, find a matrix C such that

$$AC = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 1 \end{bmatrix}.$$

17. Let

$$A^{-1} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 4 & 1 & 2 \end{bmatrix}.$$

If possible, find a matrix C such that

$$ACA = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 2 \\ 2 & 1 & 4 \end{bmatrix}.$$

18. Let

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

If possible, find a matrix B such that $AB = 2A$.

19. Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}.$$

If possible, find a matrix B such that $AB = A^2 + 2A$.20. Find all numbers r such that

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & r & 3 \\ 1 & 2 & 1 \end{bmatrix}$$

is invertible.

21. Find all numbers r such that

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & r & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

is invertible.

22. Let A and B be two $m \times n$ matrices. Show that A and B are row equivalent if and only if there exists an invertible $m \times m$ matrix C such that $CA = B$.

23. Mark each of the following True or False. The statements involve matrices A , B , and C , which are assumed to be of appropriate size.

- ___ a. If $AC = BC$ and C is invertible, then $A = B$.
- ___ b. If $AB = O$ and B is invertible, then $A = O$.
- ___ c. If $AB = C$ and two of the matrices are invertible, then so is the third.
- ___ d. If $AB = C$ and two of the matrices are singular, then so is the third.
- ___ e. If A^2 is invertible, then A^3 is invertible.
- ___ f. If A^3 is invertible, then A^2 is invertible.
- ___ g. Every elementary matrix is invertible.
- ___ h. Every invertible matrix is an elementary matrix.
- ___ i. If A and B are invertible matrices, then so is $A + B$, and $(A + B)^{-1} = A^{-1} + B^{-1}$.
- ___ j. If A and B are invertible, then so is AB , and $(AB)^{-1} = A^{-1}B^{-1}$.

24. Show that, if A is an invertible $n \times n$ matrix, then A^T is invertible. Describe $(A^T)^{-1}$ in terms of A^{-1} .

- 25. a. If A is invertible, is $A + A^T$ always invertible?
- b. If A is invertible, is $A + A$ always invertible?

26. Let A be a matrix such that A^2 is invertible. Prove that A is invertible.

27. Let A and B be $n \times n$ matrices with A invertible.

- a. Show that $AX = B$ has the unique solution $X = A^{-1}B$.
- b. Show that $X = A^{-1}B$ can be found by the following row reduction:

$$[A \mid B] \sim [I \mid X].$$

That is, if the matrix A is reduced to the identity matrix I , then the matrix B will be reduced to $A^{-1}B$.

28. Note that

$$\frac{1}{a} + \frac{1}{b} = \frac{(a+b)}{(ab)}$$

for nonzero scalars $a, b \in \mathbb{R}$. Find an analogous equality for invertible $n \times n$ matrices A and B .

29. An $n \times n$ matrix A is **nilpotent** if $A^r = O$ (the $n \times n$ zero matrix) for some positive integer r .
- Give an example of a nonzero nilpotent 2×2 matrix.
 - Show that, if A is an invertible $n \times n$ matrix, then A is not nilpotent.
30. A square matrix A is said to be **idempotent** if $A^2 = A$.
- Give an example of an idempotent matrix other than O and I .
 - Show that, if a matrix A is both idempotent and invertible, then $A = I$.
31. Show that

$$\begin{bmatrix} 0 & a_1 & a_2 & a_3 \\ 0 & 0 & b_1 & b_2 \\ 0 & 0 & 0 & c_1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

is nilpotent. (See Exercise 29.)

32. A square matrix is **upper triangular** if all entries below the main diagonal are zero. **Lower triangular** is defined symmetrically. Give an example of a nilpotent 4×4 matrix that is not upper or lower triangular. (See Exercises 29 and 31.)
33. Give an example of two invertible 4×4 matrices whose sum is singular.
34. Give an example of two singular 3×3 matrices whose sum is invertible.
35. Consider the 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

and let $h = ad - bc$.

- a. Show that, if $h \neq 0$, then

$$\begin{bmatrix} d/h & -b/h \\ -c/h & a/h \end{bmatrix}$$

is the inverse of A .

- b. Show that A is invertible if and only if $h \neq 0$.

Exercises 36–38 develop elementary column operations.

36. For each type of elementary matrix E , explain how E can be obtained from the identity matrix by means of operations on columns.
37. Let A be a square matrix, and let E be an elementary matrix of the same size. Find the effect on A of multiplying A on the right by E . [HINT: Use Exercise 36.]
38. Let A be an invertible square matrix. Recall that $(BA)^{-1} = A^{-1}B^{-1}$, and use Exercise 37 to answer the following questions:
- If two rows of A are interchanged, how does the inverse of the resulting matrix compare with A^{-1} ?
 - Answer the question in part (a) if, instead, a row of A is multiplied by a nonzero scalar r .
 - Answer the question in part (a) if, instead, r times the i th row of A is added to the j th row.



In Exercises 39–42, use the routine YUREDUCE in LINTEK to find the inverse of the matrix, if it exists. If a printer is available, make a copy of the results. Otherwise, copy down the answers to three significant figures.

39. $\begin{bmatrix} 3 & -1 & 2 \\ 1 & 2 & 1 \\ 0 & 3 & -4 \end{bmatrix}$

40. $\begin{bmatrix} -2 & 1 & 4 \\ 3 & 6 & 7 \\ 13 & 15 & -2 \end{bmatrix}$

41. $\begin{bmatrix} 2 & -1 & 3 & 4 \\ -5 & 2 & 0 & 11 \\ 12 & 13 & -6 & 8 \\ 18 & -10 & 3 & 0 \end{bmatrix}$

42. $\begin{bmatrix} 4 & -10 & 3 & 17 \\ 2 & 0 & -3 & 11 \\ 14 & 2 & 12 & -15 \\ 0 & -10 & 9 & -5 \end{bmatrix}$

In Exercises 43–48, follow the instructions for Exercises 39–42, but use the routine *MATCOMP* in *LINTEK*. Check to ensure that $AA^{-1} = I$ for each matrix A whose inverse is found.

43. The matrix in Exercise 9
44. The matrix in Exercise 10
45. The matrix in Exercise 41
46. The matrix in Exercise 40

$$\begin{aligned}
 47. & \begin{bmatrix} 4 & 1 & -3 & 2 & 6 \\ 0 & 1 & 5 & 2 & 1 \\ 3 & 8 & -11 & 4 & 6 \\ 2 & 1 & -8 & 7 & 2 \\ 1 & 3 & -1 & 4 & 8 \end{bmatrix} \\
 48. & \begin{bmatrix} 2 & -1 & 0 & 1 & 6 \\ 3 & -1 & 2 & 4 & 6 \\ 0 & 1 & 3 & 4 & 8 \\ -1 & 1 & 1 & 1 & 8 \\ 3 & 1 & 4 & -11 & 10 \end{bmatrix}
 \end{aligned}$$

MATLAB

Access *MATLAB* and, if the data files for our text are accessible, enter `fbcl55`. Otherwise, enter these four matrices by hand. [In *MATLAB*, $\ln(x)$ is denoted by `log(x)`.]

$$\begin{aligned}
 A &= \begin{bmatrix} -2 & 3 & 2/7 \\ \pi/2 & 1 & 3.2 \\ 5 & -6 & 1.3 \end{bmatrix}, & B &= \begin{bmatrix} 3\pi & \cos 2 & 21/8 \\ \sqrt{7} & \ln 4 & 2/3 \\ \sqrt{2} & \sin 4 & 8.3 \end{bmatrix}, \\
 C &= \begin{bmatrix} -3.2 & 1.4 & 5.3 \\ 1.7 & -3.6 & 4.1 \\ 10.3 & 8.5 & -7.6 \end{bmatrix}
 \end{aligned}$$

As you work the problems, write down the entry in the 2nd row, 3rd column position of the answer, with four-significant-figure accuracy, to hand in.

Enter `help inv`, read the information, and then use the function `inv` to work problems M1 through M4.

- M1. Compute C^{-3} .
- M2. Compute $A^3B^{-2}C$.
- M3. Find the matrix X such that $XB = C$.
- M4. Find the matrix X such that $B^2XC = A$.

Enter `help /` and then `help \`, read the information, and then use `/` and `\` rather than the function `inv` to work problems M5 through M8.

- M5. Compute $A^{-1}B^2C^{-1}B$.
- M6. Compute $B^{-2}CA^{-3}B^3$.
- M7. Find the matrix X such that $CX = B^{-2}$.
- M8. Find the matrix X such that $AXC^3 = B^4$.

1.6

HOMOGENEOUS SYSTEMS, SUBSPACES,
AND BASES

The Solution Set of a Homogeneous System

A linear system $Ax = b$ is **homogeneous** if $b = 0$. A homogeneous linear system $Ax = 0$ is always consistent, because $x = 0$, the zero vector, is certainly a solution. The zero vector is called the **trivial solution**. Other solutions are **nontrivial solutions**. A homogeneous system is special in that its solution set has a self-contained algebraic structure of its own, as we now show.

THEOREM 1.13 Structure of the Solution Set of $Ax = 0$

Let $Ax = 0$ be a homogeneous linear system. If h_1 and h_2 are solutions of $Ax = 0$, then so is the linear combination $rh_1 + sh_2$ for any scalars r and s .

PROOF Let h_1 and h_2 be solutions of $Ax = 0$, so that $Ah_1 = 0$ and $Ah_2 = 0$. By the distributive and scalars-pull-through properties of matrix algebra, we have

$$\begin{aligned} A(rh_1 + sh_2) &= A(rh_1) + A(sh_2) \\ &= r(Ah_1) + s(Ah_2) \\ &= r0 + s0 = 0 \end{aligned}$$

for all scalars r and s . Thus the vector $rh_1 + sh_2$ is a solution of the system $Ax = 0$. ▲

Notice how easy it was to write down the proof of Theorem 1.13 in matrix notation. What a chore it would have been to write out the proof using equations with their subscripted variables and coefficients to denote a general $m \times n$ homogeneous system!

Although we stated and proved Theorem 1.13 for just two solutions of $Ax = 0$, either induction or the same proof using k solutions shows that:

Every linear combination of solutions of a homogeneous system $Ax = 0$ is again a solution of the system.

Subspaces

The solution set of a homogeneous system $Ax = 0$ in n unknowns is an example of a subset W of $\mathbb{R}^n$ with the property that every linear combination of vectors in W is again in W . Note that W contains all linear combinations of its

vectors if and only if it contains every sum of two of its vectors and every scalar multiple of each of its vectors. We now give a formal definition of a subset of $\mathbb{R}^n$ having such a self-contained algebraic structure. Rather than phrase the definition in terms of linear combinations, we state it in terms of the two basic vector operations, vector addition and scalar multiplication.

DEFINITION 1.16 Closure and Subspace

A subset W of $\mathbb{R}^n$ is **closed under vector addition** if for all $\mathbf{u}, \mathbf{v} \in W$ the sum $\mathbf{u} + \mathbf{v}$ is in W . If $r\mathbf{v} \in W$ for all $\mathbf{v} \in W$ and all scalars r , then W is **closed under scalar multiplication**. A nonempty subset W of $\mathbb{R}^n$ that is closed under both vector addition and scalar multiplication is a **subspace** of $\mathbb{R}^n$.

Theorem 1.13 shows that the solution set of every homogeneous system with n unknowns is a subspace of $\mathbb{R}^n$. We give an example of a subset of $\mathbb{R}^2$ that is a subspace and an example of a subset that is not a subspace.

EXAMPLE 1 Show that $W = \{[x, 2x] \mid x \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.

SOLUTION Of course, W is a nonempty subset of $\mathbb{R}^2$. Let $\mathbf{u}, \mathbf{v} \in W$ so that $\mathbf{u} = [a, 2a]$ and $\mathbf{v} = [b, 2b]$. Then $\mathbf{u} + \mathbf{v} = [a, 2a] + [b, 2b] = [a + b, 2(a + b)]$ is again of the form $[x, 2x]$, and consequently is in W . This shows that W is closed under vector addition. Because $c\mathbf{u} = c[a, 2a] = [ca, 2(ca)]$ is in W , we see that W is also closed under scalar multiplication, so W is a subspace of $\mathbb{R}^2$. ■

You might recognize the subspace W of Example 1 as a line passing through the origin. However, not all lines in $\mathbb{R}^2$ are subspaces. (See Exercises 11 and 14.)

EXAMPLE 2 Determine whether $W = \{[x, y] \in \mathbb{R}^2 \mid xy \geq 0\}$ is a subspace of $\mathbb{R}^2$.

SOLUTION Here W consists of the vectors in the first or third quadrants (including the coordinate axes), as shown in Figure 1.35. As the figure illustrates, the sum of a vector in the first quadrant and a vector in the third quadrant may be a vector in the second quadrant, so W is not closed under vector addition, and is not a subspace. For a numerical example, $[1, 2] + [-2, -1] = [-1, 1]$, which is not in W . ■

Note that the set $\{\mathbf{0}\}$ consisting of just the zero vector in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$, because $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $r\mathbf{0} = \mathbf{0}$ for all scalars r . We refer to $\{\mathbf{0}\}$ as the **zero subspace**. Of course, $\mathbb{R}^n$ itself is a subspace of $\mathbb{R}^n$, because it is closed under vector addition and scalar multiplication. The two subspaces $\{\mathbf{0}\}$ and $\mathbb{R}^n$ represent extremes in size for subspaces of $\mathbb{R}^n$. The next theorem shows one way to form subspaces of various sizes.

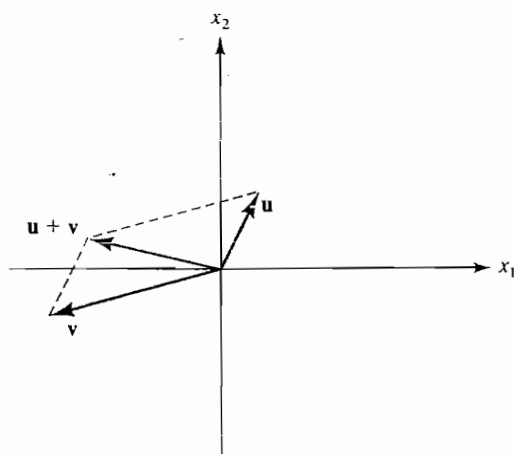


FIGURE 1.35

The shaded subset is not closed under addition.

THEOREM 1.14 Subspace Property of a Span

Let $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ be the span of $k > 0$ vectors in $\mathbb{R}^n$. Then W is a subspace of $\mathbb{R}^n$.

PROOF Let

$$\mathbf{u} = r_1\mathbf{w}_1 + r_2\mathbf{w}_2 + \cdots + r_k\mathbf{w}_k \quad \text{and} \quad \mathbf{v} = s_1\mathbf{w}_1 + s_2\mathbf{w}_2 + \cdots + s_k\mathbf{w}_k$$

be two elements of W . Their sum is

$$\mathbf{u} + \mathbf{v} = (r_1 + s_1)\mathbf{w}_1 + (r_2 + s_2)\mathbf{w}_2 + \cdots + (r_k + s_k)\mathbf{w}_k,$$

which is again a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$, so $\mathbf{u} + \mathbf{v}$ is in W . Thus W is closed under vector addition. Similarly, for any scalar c ,

$$c\mathbf{u} = (cr_1)\mathbf{w}_1 + (cr_2)\mathbf{w}_2 + \cdots + (cr_k)\mathbf{w}_k$$

is again in W —that is, W is closed under scalar multiplication. Because $k > 0$, W is also nonempty, so W is a subspace of $\mathbb{R}^n$. $\blacktriangle$

We say that the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$ **span** or **generate** the subspace $\text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ of $\mathbb{R}^n$.

We will see in Section 2.1 that every subspace in $\mathbb{R}^n$ can be described as the span of at most n vectors in $\mathbb{R}^n$. In particular, the solution set of a homogeneous system $A\mathbf{x} = \mathbf{0}$ can always be described as a span of some of the solution vectors. We illustrate how to describe the solution set this way in an example.

EXAMPLE 3 Express the solution set of the homogeneous system

$$\begin{aligned}x_1 - 2x_2 + x_3 - x_4 &= 0 \\2x_1 - 3x_2 + 4x_3 - 3x_4 &= 0 \\3x_1 - 5x_2 + 5x_3 - 4x_4 &= 0 \\-x_1 + x_2 - 3x_3 + 2x_4 &= 0\end{aligned}$$

as a span of solution vectors.

SOLUTION We reduce the augmented matrix $[A \mid \mathbf{0}]$ to transform the coefficient matrix A of the given system into reduced row-echelon form. We have

$$\left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 0 \\ 2 & -3 & 4 & -3 & 0 \\ 3 & -5 & 5 & -4 & 0 \\ -1 & 1 & -3 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & -1 & -2 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 5 & -3 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

The reduction is complete. Notice that we didn't really need to insert the column vector $\mathbf{0}$ in the augmented matrix, because it never changes.

From the reduced matrix, we find that the homogeneous system has two free variables and has a solution set described by the *general solution vector*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -5r + 3s \\ -2r + s \\ r \\ s \end{bmatrix} = r \begin{bmatrix} -5 \\ -2 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix}. \quad (1)$$

Thus the solution set is

$$\text{sp} \left(\begin{bmatrix} -5 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right).$$

We chose these two generating vectors from Eq. (1) by taking $r = 1, s = 0$ for the first and $r = 0, s = 1$ for the second. ■

The preceding example indicates how we can always express the entire solution set of a homogeneous system with k free variables as the span of k solution vectors.

Given an $m \times n$ matrix A , there are three natural subspaces of $\mathbb{R}^m$ or $\mathbb{R}^n$ associated with it. Recall (Theorem 1.14) that a span of vectors in $\mathbb{R}^n$ is always a subspace of $\mathbb{R}^n$. The span of the row vectors of A is the **row space** of A , and is of course a subspace of $\mathbb{R}^n$. The span of the column vectors of A is the **column space** of A and is a subspace of $\mathbb{R}^m$. The solution set of $A\mathbf{x} = \mathbf{0}$, which we have been discussing, is the **nullspace** of A and is a subspace of $\mathbb{R}^n$. For example, if

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \end{bmatrix},$$

we see that

the row space of A is $\text{sp}([1, 0, 3], [0, 1, -1])$ in $\mathbb{R}^3$,

the column space of A is $\text{sp}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix}\right)$ in $\mathbb{R}^2$, and

the nullspace of A is $\text{sp}\left(\begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}\right)$ in $\mathbb{R}^3$.

The nullspace of A was readily found because A is in reduced row-echelon form.

In Section 1.3, we emphasized that for an $m \times n$ matrix A and $\mathbf{x} \in \mathbb{R}^n$, the vector $A\mathbf{x}$ is a linear combination of the column vectors of A . In Section 1.4 we saw that the system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is equal to some linear combination of the column vectors of A . We rephrase this criterion for existence of a solution of $A\mathbf{x} = \mathbf{b}$ in terms of the column space of A .

Column Space Criterion

A linear system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is in the column space of A .

We have discussed the significance of the nullspace and the column space of a matrix. The row space of A is significant because the row vectors of A are orthogonal to the vectors in the nullspace of A , as the i th equation in the system $A\mathbf{x} = \mathbf{0}$ shows. This observation will be useful when we compute projections in Section 6.1.

Bases

We have seen how the solution set of a homogeneous linear system can be expressed as the span of certain selected solution vectors. Look again at Eq. (1), which shows the solution set of the linear system in Example 3 to be $\text{sp}(\mathbf{w}_1, \mathbf{w}_2)$ for

$$\mathbf{w}_1 = \begin{bmatrix} -5 \\ -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{w}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

The last two components of these vectors are 1, 0 for $\mathbf{w}_1$ and 0, 1 for $\mathbf{w}_2$. These components mirror the vectors $\mathbf{i} = [1, 0]$ and $\mathbf{j} = [0, 1]$ in the plane. Now the vector $[r, s]$ in the plane can be expressed *uniquely* as a linear combination of $\mathbf{i}$ and $\mathbf{j}$ —namely, as $r\mathbf{i} + s\mathbf{j}$. Thus we see that every solution vector in Eq. (1) of the linear system in Example 3 is a *unique* linear combination of $\mathbf{w}_1$ and $\mathbf{w}_2$ —namely, $r\mathbf{w}_1 + s\mathbf{w}_2$. We can think of (r, s) as being *coordinates* of the

solution relative to $\mathbf{w}_1$ and $\mathbf{w}_2$. Because we regard all ordered pairs of numbers as filling a plane, this indicates how we might regard the solution set of this system as a plane in $\mathbb{R}^4$. We can think of $\{\mathbf{w}_1, \mathbf{w}_2\}$ as a set of reference vectors for the plane. We have depicted this plane in Figure 1.36

More generally, if every vector $\mathbf{w}$ in the subspace $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ of $\mathbb{R}^n$ can be expressed as a linear combination $\mathbf{w} = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + \dots + c_k\mathbf{w}_k$ in a *unique* way, then we can consider the ordered k -tuple $(c_1, c_2, \dots, c_k)$ in $\mathbb{R}^k$ to be the *coordinates* of $\mathbf{w}$. The set $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ is considered to be a set of reference vectors for the subspace W . Such a set is known as a *basis* for W , as the next definition indicates.

DEFINITION 1.17 Basis for a Subspace

Let W be a subspace of $\mathbb{R}^n$. A subset $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ of W is a **basis** for W if every vector in W can be expressed *uniquely* as a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$.

Our discussion following Example 3 shows that the two vectors

$$\mathbf{w}_1 = \begin{bmatrix} -5 \\ -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{w}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

form a basis for the solution space of the homogeneous system there.

If $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ is a basis for W , then we have $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ as well as the uniqueness requirement. Remember that we called $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ *standard basis vectors* for $\mathbb{R}^n$. The reason for this is that every element of $\mathbb{R}^n$ can be expressed *uniquely* as a linear combination of these vectors $\mathbf{e}_i$. We call $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ the **standard basis** for $\mathbb{R}^n$.

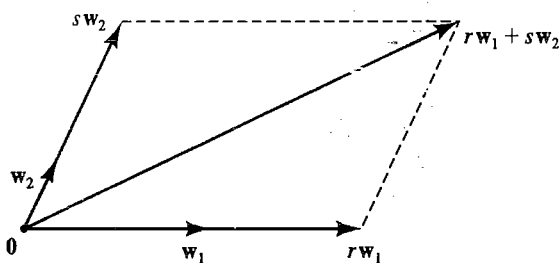


FIGURE 1.36
The plane $\text{sp}(\mathbf{w}_1, \mathbf{w}_2)$

We would like to be able to determine whether $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ is a basis for the subspace $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ of $\mathbb{R}^n$ —that is, whether the *uniqueness* criterion holds. The next theorem will be helpful. It shows that we need only examine uniqueness at the zero vector.

THEOREM 1.15 Unique Linear Combinations

The set $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ is a basis for $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ in $\mathbb{R}^n$ if and only if the zero vector is a *unique* linear combination of the $\mathbf{w}_i$ —that is, if and only if $r_1\mathbf{w}_1 + r_2\mathbf{w}_2 + \dots + r_k\mathbf{w}_k = \mathbf{0}$ implies that $r_1 = r_2 = \dots = r_k = 0$.

PROOF If $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ is a basis for W , then the expression for every vector in W as a linear combination of the $\mathbf{w}_i$ is unique, so, in particular, the linear combination that gives the zero vector must be unique. Because $0\mathbf{w}_1 + 0\mathbf{w}_2 + \dots + 0\mathbf{w}_k = \mathbf{0}$, it follows that $r_1\mathbf{w}_1 + r_2\mathbf{w}_2 + \dots + r_k\mathbf{w}_k = \mathbf{0}$ implies that each r_i must be 0.

Conversely, suppose that $0\mathbf{w}_1 + 0\mathbf{w}_2 + \dots + 0\mathbf{w}_k$ is the only linear combination giving the zero vector. If we have two linear combinations

$$\begin{aligned}\mathbf{w} &= c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + \dots + c_k\mathbf{w}_k \\ \mathbf{w} &= d_1\mathbf{w}_1 + d_2\mathbf{w}_2 + \dots + d_k\mathbf{w}_k\end{aligned}$$

for a vector $\mathbf{w} \in W$, then, subtracting these two equations, we obtain

$$\mathbf{0} = (c_1 - d_1)\mathbf{w}_1 + (c_2 - d_2)\mathbf{w}_2 + \dots + (c_k - d_k)\mathbf{w}_k.$$

From the unique linear combination giving the zero vector, we see that

$$c_1 - d_1 = c_2 - d_2 = \dots = c_k - d_k = 0,$$

and so $c_i = d_i$ for $i = 1, 2, \dots, k$, showing that the linear combination giving $\mathbf{w}$ is unique. $\blacktriangle$

The Unique Solution Case for $A\mathbf{x} = \mathbf{b}$

The preceding theorem immediately focuses our attention on determining when a linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution. Our boxed column space criterion asserts that the system $A\mathbf{x} = \mathbf{b}$ has *at least one* solution precisely when $\mathbf{b}$ is in the column space of A . By Definition 1.17, the system has *exactly one* solution for each $\mathbf{b}$ in the column space of A if and only if the column vectors of A form a basis for this column space.

Let A be a square $n \times n$ matrix. Then each column vector $\mathbf{b}$ in $\mathbb{R}^n$ is a unique linear combination of the column vectors of A if and only if $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$. By Theorem 1.12, this is equivalent to A being row-reducible to the identity matrix, so that A is invertible. We have now established another equivalent condition to add to those in Theorem 1.12. We summarize the main ones in a new theorem for easy reference.

THEOREM 1.16 The Square Case, $m = n$

Let A be an $n \times n$ matrix. The following are equivalent.

1. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$.
2. The matrix A is row equivalent to the identity matrix I .
3. The matrix A is invertible.
4. The column vectors of A form a basis for $\mathbb{R}^n$.

In describing examples and stating exercises, we often write vectors in $\mathbb{R}^n$ as row vectors to save space. However, the vector $\mathbf{b} = A\mathbf{x}$ in Theorem 1.16 is necessarily a column vector. Thus, solutions to examples and exercises that use results such as Theorem 1.16 are done using column-vector notation, as our next example illustrates.

EXAMPLE 4 Determine whether the vectors $\mathbf{v}_1 = [1, 1, 3]$, $\mathbf{v}_2 = [3, 0, 4]$, and $\mathbf{v}_3 = [1, 4, -1]$ form a basis for $\mathbb{R}^3$.

SOLUTION We must see whether the matrix A having $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ as column vectors is row equivalent to the identity matrix. We need only create zeros below pivots to determine if this is the case. We obtain

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 0 & 4 \\ 3 & 4 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & -3 & 3 \\ 0 & -5 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -9 \end{bmatrix}.$$

There is no point in going further. We see that we will be able to get the identity matrix, so $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$. ■

A linear system having the same number n of equations as unknowns is called a **square system**, because the coefficient matrix is a square $n \times n$ matrix. When a square matrix is reduced to echelon form, the result is a square matrix having only zero entries below the *main diagonal*, which runs from the upper left-hand corner to the lower right-hand corner. This follows at once from the fact that the pivot in a nonzero row—say, the i th row—is always in a column j , where $j \geq i$. Such a square matrix U with zero entries below the main diagonal is called **upper triangular**. The final matrix displayed in Example 4 is upper triangular.

For a general linear system $A\mathbf{x} = \mathbf{b}$ of m equations in n unknowns, we consult Theorem 1.7. It tells us that a consistent system $A\mathbf{x} = \mathbf{b}$ has a *unique* solution if and only if a row-echelon form H of A has a pivot in each of its n columns. Because no two pivots appear in the same row of H , we see that H has at least as many rows as columns; that is, $m \geq n$. Consequently, the reduced row-echelon form for A must consist of the identity matrix, followed by $m - n$

zero rows. For example, if $m = 5$ and $n = 3$, the reduced row-echelon form for A in this unique solution case must be

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We summarize these observations as a theorem.

THEOREM 1.17 The General Unique Solution Case

Let A be an $m \times n$ matrix. The following are equivalent.

1. Each consistent system $Ax = b$ has a unique solution.
2. The reduced row-echelon form of A consists of the $n \times n$ identity matrix followed by $m - n$ rows of zeros.
3. The column vectors of A form a basis for the column space of A .

EXAMPLE 5 Determine whether the vectors $w_1 = [1, 2, 3, -1]$, $w_2 = [-2, -3, -5, 1]$, and $w_3 = [-1, -3, -4, 2]$ form a basis for the subspace $\text{sp}(w_1, w_2, w_3)$ in $\mathbb{R}^4$.

SOLUTION By Theorem 1.17, we need to determine whether the reduced row-echelon form of the matrix A with w_1 , w_2 , and w_3 as column vectors consists of the 3×3 identity matrix followed by a row of zeros. Again, we can determine this using just the row-echelon form, without creating zeros above the pivots. We obtain

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -3 & -3 \\ 3 & -5 & -4 \\ -1 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We cannot obtain the 3×3 identity matrix. Thus the vectors do not form a basis for the subspace which is the column space of A . ■

A linear system having an infinite number of solutions is called **underdetermined**. We now prove a corollary of the preceding theorem: that a consistent system is underdetermined if it has fewer equations than unknowns.

COROLLARY 1 Fewer Equations than Unknowns, $m < n$

If a linear system $Ax = b$ is consistent and has fewer equations than unknowns, then it has an infinite number of solutions.

PROOF If $m < n$ in Theorem 1.17, the reduced row-echelon form of A cannot contain the $n \times n$ identity matrix, so we cannot be in the unique solution case. Because we are assuming that the system is consistent, there are an infinite number of solutions. $\blacktriangle$

The next corollary follows at once from Corollary 1 and Theorem 1.17.

COROLLARY 2 The Homogeneous Case

1. A homogeneous linear system $Ax = 0$ having fewer equations than unknowns has a nontrivial solution—that is, a solution other than the zero vector.
 2. A square homogeneous system $Ax = 0$ has a nontrivial solution if and only if A is not row equivalent to the identity matrix of the same size.
-

EXAMPLE 6 Show that a basis for $\mathbb{R}^n$ cannot contain more than n vectors.

SOLUTION If $\{v_1, v_2, \dots, v_k\}$ is a basis for $\mathbb{R}^n$, then, by Theorem 1.15, the only linear combination of the v_j equal to the zero vector is the one for which the coefficient of each v_j is the scalar 0. In terms of a matrix equation, the homogeneous linear system $Ax = 0$, where v_j is the j th column vector of the $n \times k$ matrix A , must have only the trivial solution. If $k > n$, then this linear system has fewer equations than unknowns, and therefore a nontrivial solution by Corollary 2. Consequently, we must have $k \leq n$. $\blacksquare$

The Solution Set of $Ax = b$

Theorem 1.13 tells us the structure of the solution set of a homogeneous linear system. To conclude the section, we now describe the solution set of $Ax = b$ in terms of the solution set of the *corresponding homogeneous system* $Ax = 0$. It is customary to refer to an equation that describes the whole solution set as the **general solution**, and to refer to each element of the solution set as a **particular solution**.

THEOREM 1.18 Structure of the Solution Set of $Ax = b$

Let $Ax = b$ be a linear system. If p is any particular solution of $Ax = b$ and h is a solution of the corresponding homogeneous system $Ax = 0$, then $p + h$ is a solution of $Ax = b$. Moreover, every solution of $Ax = b$ has this form $p + h$, so that the general solution is $x = p + h$ where $Ah = 0$.

PROOF Let p be a solution of $Ax = b$, so that $Ap = b$, and let h be a solution of $Ax = 0$, so that $Ah = 0$. Then

$$A(p + h) = Ap + Ah = b + 0 = b,$$

and so $\mathbf{p} + \mathbf{h}$ is indeed a solution. Moreover, if $\mathbf{q}$ is any solution of $A\mathbf{x} = \mathbf{b}$, then

$$A(\mathbf{q} - \mathbf{p}) = A\mathbf{q} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0},$$

and so $\mathbf{q} - \mathbf{p}$ is a solution $\mathbf{h}$ of $A\mathbf{x} = \mathbf{0}$. From $\mathbf{q} - \mathbf{p} = \mathbf{h}$, it follows that $\mathbf{q} = \mathbf{p} + \mathbf{h}$. This completes the proof. $\blacktriangle$

Students who have studied differential equations may be familiar with a similar theorem describing the general solution of a linear differential equation.

EXAMPLE 7 Illustrate Theorem 1.18 for the linear system $A\mathbf{x} = \mathbf{b}$ given by

$$x_1 - 2x_2 + x_3 - x_4 = 4$$

$$2x_1 - 3x_2 + 4x_3 - 3x_4 = -1$$

$$3x_1 - 5x_2 + 5x_3 - 4x_4 = 3$$

$$-x_1 + x_2 - 3x_3 + 2x_4 = 5.$$

SOLUTION We reduce the augmented matrix $[A \mid \mathbf{b}]$ to transform the coefficient matrix A of the given system into reduced row-echelon form. (The coefficient matrix A is the same as in Example 3.) We have

$$\left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 4 \\ 2 & -3 & 4 & -3 & -1 \\ 3 & -5 & 5 & -4 & 3 \\ -1 & 1 & -3 & 2 & 5 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 4 \\ 0 & 1 & 2 & -1 & -9 \\ 0 & 1 & 2 & -1 & -9 \\ 0 & -1 & -2 & 1 & 9 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 5 & -3 & -14 \\ 0 & 1 & 2 & -1 & -9 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Writing the general solution in the usual form and then as described in Theorem 1.18, we have

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -14 - 5r + 3s \\ -9 - 2r + s \\ r \\ s \end{bmatrix} = \begin{bmatrix} -14 \\ -9 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -5r + 3s \\ -2r + s \\ r \\ s \end{bmatrix}.$$

General solution $\mathbf{p}$ $\mathbf{h}$

SUMMARY

1. A linear system $A\mathbf{x} = \mathbf{b}$ is *homogeneous* if $\mathbf{b} = \mathbf{0}$.
2. Every linear combination of solutions of a homogeneous system $A\mathbf{x} = \mathbf{0}$ is again a solution of the system.
3. A subset W of $\mathbb{R}^n$ is *closed under vector addition* if the sum of two vectors in W is again in W . The subset W is *closed under scalar multiplication* if every scalar multiple of every vector in W is in W . If W is nonempty and closed under both operations, then W is a *subspace* of $\mathbb{R}^n$.

- The span of any k vectors in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$. If A is an $m \times n$ matrix, the *row space* of A is the span in $\mathbb{R}^n$ of the row vectors of A , the *column space* of A is the span in $\mathbb{R}^m$ of the column vectors, and the *nullspace* of A is the solution set of $Ax = 0$ in $\mathbb{R}^n$.
- A subset $\{w_1, w_2, \dots, w_k\}$ of a subspace W of $\mathbb{R}^n$ is a *basis* for W if every vector in W can be expressed uniquely as a linear combination of $w_1, w_2, \dots, w_k$.
- The set $\{w_1, w_2, \dots, w_k\}$ is a basis for $\text{sp}(w_1, w_2, \dots, w_k)$ if and only if $0w_1 + 0w_2 + \dots + 0w_k$ is the unique linear combination of the w_i that is equal to the zero vector.
- A consistent linear system $Ax = b$ of m equations in n unknowns has a unique solution if and only if the reduced row-echelon form of A appears as the $n \times n$ identity matrix followed by $m - n$ rows of zeros.
- A consistent linear system having fewer equations than unknowns is underdetermined—that is, it has an infinite number of solutions.
- A square linear system has a unique solution if and only if its coefficient matrix is row equivalent to the identity matrix.
- The solutions of any consistent linear system $Ax = b$ are precisely the vectors $p + h$, where p is any one particular solution of $Ax = b$ and h varies through the solution set of the homogeneous system $Ax = 0$.

EXERCISES

In Exercises 1–10, determine whether the indicated subset is a subspace of the given Euclidean space $\mathbb{R}^n$.

- $\{[r, -r] \mid r \in \mathbb{R}\}$ in $\mathbb{R}^2$
- $\{[x, x + 1] \mid x \in \mathbb{R}\}$ in $\mathbb{R}^2$
- $\{[n, m] \mid n \text{ and } m \text{ are integers}\}$ in $\mathbb{R}^2$
- $\{[x, y] \mid x, y \in \mathbb{R} \text{ and } x, y \geq 0\}$ (the first quadrant of $\mathbb{R}^2$)
- $\{[x, y, z] \mid x, y, z \in \mathbb{R} \text{ and } z = 3x + 2\}$ in $\mathbb{R}^3$
- $\{[x, y, z] \mid x, y, z \in \mathbb{R} \text{ and } x = 2y + z\}$ in $\mathbb{R}^3$
- $\{[x, y, z] \mid x, y, z \in \mathbb{R} \text{ and } z = 1, y = 2x\}$ in $\mathbb{R}^3$
- $\{[2x, x + y, y] \mid x, y \in \mathbb{R}\}$ in $\mathbb{R}^3$
- $\{[2x_1, 3x_2, 4x_3, 5x_4] \mid x_i \in \mathbb{R}\}$ in $\mathbb{R}^4$
- $\{[x_1, x_2, \dots, x_n] \mid x_i \in \mathbb{R}, x_2 = 0\}$ in $\mathbb{R}^n$
- Prove that the line $y = mx$ is a subspace of $\mathbb{R}^2$. [HINT: Write the line as $W = \{[x, mx] \mid x \in \mathbb{R}\}$.]
- Let a, b , and c be scalars such that $abc \neq 0$. Prove that the plane $ax + by + cz = 0$ is a subspace of $\mathbb{R}^3$.
- Give a geometric description of all subspaces of $\mathbb{R}^3$.
 - Repeat part (a) for $\mathbb{R}^3$.
- Prove that every subspace of $\mathbb{R}^n$ contains the zero vector.
- Is the zero vector a basis for the subspace $\{0\}$ of $\mathbb{R}^n$? Why or why not?

In Exercises 16–21, find a basis for the solution set of the given homogeneous linear system.

- $$\begin{aligned} x - y &= 0 \\ 2x - 2y &= 0 \end{aligned}$$
- $$\begin{aligned} 3x_1 + x_2 + x_3 &= 0 \\ 6x_1 + 2x_2 + 2x_3 &= 0 \\ -9x_1 - 3x_2 - 3x_3 &= 0 \end{aligned}$$

18. $x_1 - x_2 + x_3 - x_4 = 0$

$x_2 + x_3 = 0$

$x_1 + 2x_2 - x_3 + 3x_4 = 0$

19. $2x_1 + x_2 + x_3 + x_4 = 0$

$x_1 - 6x_2 + x_3 = 0$

$3x_1 - 5x_2 + 2x_3 + x_4 = 0$

$5x_1 - 4x_2 + 3x_3 + 2x_4 = 0$

20. $2x_1 + x_2 + x_3 + x_4 = 0$

$3x_1 + x_2 - x_3 + 2x_4 = 0$

$x_1 + x_2 + 3x_3 = 0$

$x_1 - x_2 - 7x_3 + 2x_4 = 0$

21. $x_1 - x_2 + 6x_3 + x_4 - x_5 = 0$

$3x_1 + 2x_2 - 3x_3 + 2x_4 + 5x_5 = 0$

$4x_1 + 2x_2 - x_3 + 3x_4 - x_5 = 0$

$3x_1 - 2x_2 + 14x_3 + x_4 - 8x_5 = 0$

$2x_1 - x_2 + 8x_3 + 2x_4 - 7x_5 = 0$

In Exercises 22–30, determine whether the set of vectors is a basis for the subspace of $\mathbb{R}^n$ that the vectors span.

22. $\{[-1, 1], [1, 2]\}$ in $\mathbb{R}^2$

23. $\{[-1, 3, 1], [2, 1, 4]\}$ in $\mathbb{R}^3$

24. $\{[-1, 3, 4], [1, 5, -1], [1, 13, 2]\}$ in $\mathbb{R}^3$

25. $\{[2, 1, -3], [4, 0, 2], [2, -1, 3]\}$ in $\mathbb{R}^3$

26. $\{[2, 1, 0, 2], [2, -3, 1, 0], [3, 2, 0, 0]\}$ in $\mathbb{R}^4$

27. The set of row vectors of the matrix

$$\begin{bmatrix} 2 & -6 & 1 \\ 1 & -3 & 4 \end{bmatrix}$$

28. The set of column vectors of the matrix in Exercise 27.

29. The set of row vectors of the matrix

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & -3 \\ 1 & -3 & 4 \end{bmatrix}$$

30. The set of column vectors of the matrix in Exercise 29.

31. Find a basis for the nullspace of the matrix

$$\begin{bmatrix} 2 & 3 & 1 \\ 5 & 2 & 1 \\ 1 & 7 & 2 \\ 6 & -2 & 0 \end{bmatrix}$$

32. Find a basis for the nullspace of the matrix

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 0 & 4 & 2 \\ 3 & 2 & 8 & 7 \end{bmatrix}$$

33. Let $v_1, v_2, \dots, v_k$ and $w_1, w_2, \dots, w_m$ be vectors in a vector space V . Give a necessary and sufficient condition, involving linear combinations, for

$$\text{sp}(v_1, v_2, \dots, v_k) = \text{sp}(w_1, w_2, \dots, w_m).$$

In Exercises 34–37, solve the given linear system and express the solution set in a form that illustrates Theorem 1.18.

34. $x_1 - 2x_2 + x_3 + 5x_4 = 7$

35. $2x_1 - x_2 + 3x_3 = -3$

$4x_1 + 2x_2 - x_4 = 1$

36. $x_1 - 2x_2 + x_3 + x_4 = 4$

$2x_1 + x_2 - 3x_3 - x_4 = 6$

$x_1 - 7x_2 - 6x_3 + 2x_4 = 6$

37. $2x_1 + x_2 + 3x_3 = 5$

$x_1 - x_2 + 2x_3 + x_4 = 0$

$4x_1 - x_2 + 7x_3 + 2x_4 = 5$

$-x_1 - 2x_2 - x_3 + x_4 = -5$

38. Mark each of the following True or False.

- a. A linear system with fewer equations than unknowns has an infinite number of solutions.
- b. A consistent linear system with fewer equations than unknowns has an infinite number of solutions.
- c. If a square linear system $Ax = b$ has a solution for every choice of column vector b , then the solution is unique for each b .
- d. If a square system $Ax = 0$ has only the trivial solution, then $Ax = b$ has a unique solution for every column vector b with the appropriate number of components.
- e. If a linear system $Ax = 0$ has only the trivial solution, then $Ax = b$ has a unique solution for every column vector b with the appropriate number of components.
- f. The sum of two solution vectors of any linear system is also a solution vector of the system.

- g. The sum of two solution vectors of any homogeneous linear system is also a solution vector of the system.
 - h. A scalar multiple of a solution vector of any homogeneous linear system is also a solution vector of the system.
 - i. Every line in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$ generated by a single vector.
 - j. Every line through the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$ generated by a single vector.
39. We have defined a linear system to be *underdetermined* if it has an infinite number of solutions. Explain why this is a reasonable term to use for such a system.
40. A linear system is *overdetermined* if it has more equations than unknowns. Explain why this is a reasonable term to use for such a system.
41. Referring to Exercises 39 and 40, give an example of an overdetermined underdetermined linear system!
42. Use Theorem 1.13 to explain why a homogeneous system of linear equations has either a unique solution or an infinite number of solutions.
43. Use Theorem 1.18 to explain why no system of linear equations can have exactly two solutions.
44. Let A be an $m \times n$ matrix such that the homogeneous system $Ax = 0$ has only the trivial solution.
- a. Does it follow that every system $Ax = b$ is consistent?
 - b. Does it follow that every consistent system $Ax = b$ has a unique solution?
45. Let v_1 and v_2 be vectors in $\mathbb{R}^n$. Prove the following set equalities by showing that each of the spans is contained in the other.
- a. $\text{sp}(v_1, v_2) = \text{sp}(v_1, 2v_1 + v_2)$
 - b. $\text{sp}(v_1, v_2) = \text{sp}(v_1 + v_2, v_1 - v_2)$
46. Referring to Exercise 45, prove that if $\{v_1, v_2\}$ is a basis for $\text{sp}(v_1, v_2)$, then
- a. $\{v_1, 2v_1 + v_2\}$ is also a basis.
 - b. $\{v_1 + v_2, v_1 - v_2\}$ is also a basis.
 - c. $\{v_1 + v_2, v_1 - v_2, 2v_1 - 3v_2\}$ is not a basis.
47. Let W_1 and W_2 be two subspaces of $\mathbb{R}^n$. Prove that their intersection $W_1 \cap W_2$ is also a subspace.
-  In Exercises 48–51, use *LINTEK* or *MATLAB* to determine whether the given vectors form a basis for the subspace of $\mathbb{R}^n$ that they span.
48. $a_1 = [1, 1, -1, 0]$
 $a_2 = [5, 1, 1, 2]$
 $a_3 = [5, -3, 2, -1]$
 $a_4 = [9, 3, 0, 3]$
49. $b_1 = [3, -4, 0, 0, 1]$
 $b_2 = [4, 0, 2, -6, 2]$
 $b_3 = [0, 1, 1, -3, 0]$
 $b_4 = [1, 4, -1, 3, 0]$
50. $v_1 = [4, -1, 2, 1]$
 $v_2 = [10, -2, 5, 1]$
 $v_3 = [-9, 1, -6, -3]$
 $v_4 = [1, -1, 0, 0]$
51. $w_1 = [1, 4, -8, 16]$
 $w_2 = [1, 1, -1, 1]$
 $w_3 = [1, 4, 8, 16]$
 $w_4 = [1, 1, 1, 1]$

MATLAB

Access *MATLAB* and enter **fbclsb6** if our text data files are available; otherwise, enter the vectors in Exercises 48–51 by hand. Use *MATLAB* matrix commands to form the necessary matrix and reduce it in problems M1–M4.

M1. Solve Exercise 48.

M3. Solve Exercise 50.

M2. Solve Exercise 49.

M4. Solve Exercise 51.

- M5. What do you think the probability would be that if n vectors in $\mathbb{R}^n$ were selected at random, they would form a basis for $\mathbb{R}^n$. (The *probability* of an event is a number from 0 to 1. An impossible event has probability 0, a certain event has probability 1, an event as likely to occur as not to occur has probability 0.5, etc.)
- M6. As a way of testing your answer to the preceding exercise, you might experiment by asking MATLAB to generate “random” $n \times n$ matrices for some value of n , and reducing them to see if their column vectors form a basis for $\mathbb{R}^n$. Enter `rand(8)` to view an 8×8 matrix with “random” entries between 0 and 1. The column vectors cannot be considered to be random vectors in $\mathbb{R}^8$, because all their components lie between 0 and 1. Do you think the probability that such column vectors form a basis for $\mathbb{R}^8$ is the same as in the preceding exercise? As an experimental check, execute the command `rref(rand(8))` ten times to row-reduce ten such 8×8 matrices, examining each reduced matrix to see if the column vectors of the matrix generated by `rand(8)` did form a basis for $\mathbb{R}^8$.
- M7. Note that `4*rand(8)-2*ones(8)` will produce an 8×8 matrix with “random” entries between -2 and 2 . Again, its column vectors cannot be regarded as random vectors in $\mathbb{R}^8$, but at least the components of the vectors need not all be positive, as they were in the preceding exercise. Do you think the probability that such column vectors form a basis for $\mathbb{R}^8$ is the same as in Exercise M5? As an experimental check, row-reduce ten such matrices.

1.7

APPLICATION TO POPULATION DISTRIBUTION (OPTIONAL)

Linear algebra has proved to be a valuable tool for many practical and mathematical problems. In this section, we present an application to population distribution (Markov chains).

Consider situations in which people are split into two or more categories. For example, we might split the citizens of the United States according to income into categories of

poor, middle income, rich.

We might split the inhabitants of North America into categories according to the climate in which they live:

hot, temperate, cold.

In this book, we will speak of a *population* split into *states*. In the two illustrations above, the populations and states are given by the following:

Population	States
Citizens of the United States	poor, middle income, rich
People in North America	hot, temperate, cold

Our populations will often consist of people, but this is not essential. For example, at any moment we can classify the population of cars as operational or not operational.

We are interested in how the distribution of a population between (or among) states may change over a period of time. Matrices and their multiplication can play an important role in such considerations.

Transition Matrices

The tendency of a population to move among n states can sometimes be described using an $n \times n$ matrix. Consider a population distributed among $n = 3$ states, which we call state 1, state 2, and state 3. Suppose that we know the proportion t_{ij} of the population of state j that moves to state i over a given fixed time period. Notice that the direction of movement from state j to state i is the right-to-left order of the subscripts in t_{ij} . The matrix $T = [t_{ij}]$ is called a **transition matrix**. (Do not confuse our use of T as a transition matrix in this one section with our use of T as a linear transformation elsewhere in the text.)

EXAMPLE 1 Let the population of a country be classified according to income as

State 1: poor,

State 2: middle income,

State 3: rich.

Suppose that, over each 20-year period (about one generation), we have the following data for people and their offspring:

Of the poor people, 19% become middle income and 1% rich.

Of the middle income people, 15% become poor and 10% rich.

Of the rich people, 5% become poor and 30% middle income.

Give the transition matrix describing these data.

SOLUTION The entry t_{ij} in the transition matrix T represents the proportion of the population moving from state j to state i , not the percentage. Thus, because 19% of the poor (state 1) will become middle income (state 2), we should take $t_{21} = .19$. Similarly, because 1% of the people in state 1 move to state 3 (rich), we should take $t_{31} = .01$. Now t_{11} represents the proportion of the poor people who remain poor at the end of 20 years. Because this is 80%, we should take $t_{11} = .80$. Continuing in this fashion, starting in state 2 and then in state 3, we obtain the matrix

$$T = \begin{array}{ccc} & \begin{array}{ccc} \text{poor} & \text{mid} & \text{rich} \end{array} \\ \begin{array}{c} \text{poor} \\ \text{mid} \\ \text{rich} \end{array} & \begin{bmatrix} .80 & .15 & .05 \\ .19 & .75 & .30 \\ .01 & .10 & .65 \end{bmatrix} \end{array}$$

We have labeled the columns and rows with the names of the states. Notice that an entry of the matrix gives the proportion of the population in the state above the entry that moves to the state at the right of the entry during one 20-year period. ■

In Example 1, the sum of the entries in each column of T is 1, because the sum reflects the movement of the entire population for the state listed at the top of the column. Now suppose that the proportions of the entire population in Example 1 that fall into the various states at the start of a time period are given in the column vector

$$\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}.$$

For example, we would have

$$\mathbf{p} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

if the whole population were initially equally divided among the states. The entries in such a **population distribution vector** $\mathbf{p}$ must be nonnegative and must have a sum equal to 1.

Let us find the proportion of the entire population that is in state 1 after one time period of 20 years, knowing that initially the proportion in state 1 is p_1 . The proportion of the state-1 population that remains in state 1 is t_{11} . This gives a contribution of $t_{11}p_1$ to the proportion of the entire population that will be found in state 1 at the end of 20 years. Of course, we also get contributions to state 1 at the end of 20 years from states 2 and 3. These two states contribute proportions $t_{12}p_2$ and $t_{13}p_3$ of the entire population to state 1. Thus, after 20 years, the proportion in state 1 is

$$t_{11}p_1 + t_{12}p_2 + t_{13}p_3.$$

This is precisely the first entry in the column vector given by the product

$$T\mathbf{p} = \begin{bmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ t_{31} & t_{32} & t_{33} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}.$$

In a similar fashion, we find that the second and third components of $T\mathbf{p}$ give the proportions of population in state 2 and in state 3 after one time period.

For an initial population distribution vector $\mathbf{p}$ and transition matrix T the product vector $T\mathbf{p}$ is the population distribution vector after one time period.

Markov Chains

In Example 1, we found the transition matrix governing the flow of a population among three states over a period of 20 years. Suppose that the same transition matrix is valid over the next 20-year period, and for the next 20 years after that, and so on. That is, suppose that there is a sequence or *chain* of 20-year periods over which the transition matrix is valid. Such a situation is called a **Markov chain**. Let us give a formal definition of a transition matrix for a Markov chain.

DEFINITION 1.18 Transition Matrix

An $n \times n$ matrix T is the **transition matrix** for an n -state Markov chain if all entries in T are nonnegative and the sum of the entries in each column of T is 1.

Markov chains arise naturally in biology, psychology, economics, and many other sciences. Thus they are an important application of linear algebra and of probability. The entry t_{ij} in a transition matrix T is known as the **probability** of moving from state j to state i over one time period.

EXAMPLE 2 Show that the matrix

$$T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

is a transition matrix for a three-state Markov chain, and explain the significance of the zeros and the ones.

SOLUTION The entries are all nonnegative, and the sum of the entries in each column is 1. Thus the matrix is a transition matrix for a Markov chain.

At least for finite populations, a transition probability $t_{ij} = 0$ means that there is no movement from state j to state i over the time period. That is,

HISTORICAL NOTE MARKOV CHAINS are named for the Russian mathematician Andrei Andreevich Markov (1856–1922), who first defined them in a paper of 1906 dealing with the Law of Large Numbers and subsequently proved many of the standard results about them. His interest in these sequences stemmed from the needs of probability theory; Markov never dealt with their applications to the sciences. The only real examples he used were from literary texts, where the two possible states were vowels and consonants. To illustrate his results, he did a statistical study of the alternation of vowels and consonants in Pushkin's *Eugene Onegin*.

Andrei Markov taught at St. Petersburg University from 1880 to 1905, when he retired to make room for younger mathematicians. Besides his work in probability, he contributed to such fields as number theory, continued fractions, and approximation theory. He was an active participant in the liberal movement in the pre-World War I era in Russia; on many occasions he made public criticisms of the actions of state authorities. In 1913, when as a member of the Academy of Sciences he was asked to participate in the pompous ceremonies celebrating the 300th anniversary of the Romanov dynasty, he instead organized a celebration of the 200th anniversary of Jacob Bernoulli's publication of the Law of Large Numbers.

transition from state j to state i over the time period is impossible. On the other hand, if $t_{ij} = 1$, the entire population of state j moves to state i over the time period. That is, transition from state j to state i in the time period is certain.

For the given matrix, we see that, over one time period, the entire population of state 1 moves to state 2, the entire population of state 2 moves to state 3, and the entire population of state 3 moves to state 1. ■

If T is an $n \times n$ transition matrix and $\mathbf{p}$ is a population distribution column vector with n components, then we can readily see that $T\mathbf{p}$ is again a population distribution vector. We illustrate the general argument with the case $n = 2$, avoiding summation notation and saving space. We have

$$T\mathbf{p} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} t_{11}p_1 + t_{12}p_2 \\ t_{21}p_1 + t_{22}p_2 \end{bmatrix}.$$

To show that the sum of the components of $T\mathbf{p}$ is 1, we simply rearrange the sum of the four products involved so that the terms involving p_1 appear first, followed by the terms involving p_2 . We obtain

$$\begin{aligned} t_{11}p_1 + t_{21}p_1 + t_{12}p_2 + t_{22}p_2 &= p_1(t_{11} + t_{21}) + p_2(t_{12} + t_{22}) \\ &= p_1(1) + p_2(1) = p_1 + p_2 = 1. \end{aligned}$$

The proof for the $n \times n$ case is identical; we would have n^2 products rather than four. Note that it follows that if T is a transition matrix, then so is T^2 ; we need only observe that the j th column $\mathbf{c}$ of T is itself a population distribution vector, so the j th column of T^2 , which is $T\mathbf{c}$, has a component sum equal to 1.

Let T be the transition matrix over a time period—say, 20 years—in a Markov chain. We can form a new Markov chain by looking at the flow of the population over a time period twice as long—that is, over 40 years. Let us see the relationship of the transition matrix for the 40-year time period to the one for the 20-year time period. We might guess that the transition matrix for 40 years is T^2 . This is indeed the case. First, note that the j th column vector of an $n \times n$ matrix A is $A\mathbf{e}_j$, where $\mathbf{e}_j$ is the j th standard basis vector of $\mathbb{R}^n$, regarded as a column vector. Now $\mathbf{e}_j$ is a population distribution vector, so the j th column of the two-period transition matrix is $T(T\mathbf{e}_j) = T^2\mathbf{e}_j$, showing that T^2 is indeed the two-period matrix.

If we extend the argument above, we find that the three-period transition matrix for the Markov chain is T^3 , and so on. This exhibits another situation in which matrix multiplication is useful. Although raising even a small matrix to a power using pencil and paper is tedious, a computer can do it easily. LINTEK and MATLAB can be used to compute a power of a matrix.

m-Period Transition Matrix

A Markov chain with transition matrix T has T^m as its m -period transition matrix.

EXAMPLE 3 For the transition matrix

$$T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

in Example 2, show that, after three time periods, the distribution of population among the three states is the same as the initial population distribution.

SOLUTION After three time periods, the population distribution vector is $T^3\mathbf{p}$, and we easily compute that $T^3 = I$, the 3×3 identity matrix. Thus $T^3\mathbf{p} = \mathbf{p}$, as asserted. Alternatively, we could note that the entire population of state 1 moves to state 2 in the first time period, then to state 3 in the next time period, and finally back to state 1 in the third time period. Similarly, the populations of the other two states move around and then back to the beginning state over the three periods. ■

Regular Markov Chains

We now turn to Markov chains where there exists some fixed number m of time periods in which it is possible to get from any state to any other state. This means that the m th power T^m of the transition matrix has no zero entries.

DEFINITION 1.19 Regular Transition Matrix, Regular Chain

A transition matrix T is **regular** if T^m has no zero entries for some integer m . A Markov chain having a regular transition matrix is called a **regular chain**.

EXAMPLE 4 Show that the transition matrix

$$T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

of Example 2 is not regular.

SOLUTION A computation shows that T^2 still has zero entries. We saw in Example 3 that $T^3 = I$, the 3×3 identity matrix, so we must have $T^4 = T$, $T^5 = T^2$, $T^6 = T^3 = I$, and the powers of T repeat in this fashion. We never eliminate all the zeros. Thus T is not a regular transition matrix. ■

If T^m has no zero entries, then $T^{m+1} = (T^m)T$ has no zero entries, because the entries in any column vector of T are nonnegative with at least one nonzero entry. In determining whether a transition matrix is regular, it is not necessary to compute the entries in powers of the matrix. We need only determine whether or not they are zero.

EXAMPLE 5 If $\times$ denotes a nonzero entry, determine whether a transition matrix T with the zero and nonzero configuration given by

$$T = \begin{bmatrix} \times & 0 & \times & 0 \\ 0 & 0 & \times & 0 \\ \times & 0 & 0 & \times \\ 0 & \times & 0 & 0 \end{bmatrix}$$

is regular.

SOLUTION We compute configurations of high powers of T as rapidly as we can, because once a power has no zero entries, all higher powers must have nonzero entries. We find that

$$T^2 = \begin{bmatrix} \times & 0 & \times & \times \\ \times & 0 & 0 & \times \\ \times & \times & \times & 0 \\ 0 & 0 & \times & 0 \end{bmatrix}, \quad T^4 = \begin{bmatrix} \times & \times & \times & \times \\ \times & 0 & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & 0 \end{bmatrix}, \quad T^8 = \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{bmatrix},$$

so the matrix T is indeed regular. ■

It can be shown that, if a Markov chain is regular, the distribution of population among the states over many time periods approaches a fixed *steady-state distribution vector* $\mathbf{s}$. That is, the distribution of population among the states no longer changes significantly as time progresses. This is not to say that there is no longer movement of population between states; the transition matrix T continues to effect changes. But the movement of population out of any state over one time period is balanced by the population moving into that state, so the proportion of the total population in that state remains constant. This is a consequence of the following theorem, whose proof is beyond the scope of this book.

THEOREM 1.19 Achievement of Steady State

Let T be a regular transition matrix. There exists a unique column vector $\mathbf{s}$ with strictly positive entries whose sum is 1 such that the following hold:

1. As m becomes larger and larger, all columns of T^m approach the column vector $\mathbf{s}$.
2. $T\mathbf{s} = \mathbf{s}$, and $\mathbf{s}$ is the unique column vector with this property and whose components add up to 1.

From Theorem 1.19 we can show that, if $\mathbf{p}$ is any initial population distribution vector for a regular Markov chain with transition matrix T , the population distribution vector after many time periods approaches the vector